SECURITY ENGINEERING RESEARCH POSTER

Evidence-Centered Visualization for ML Security Engineering

Aggregating Repository-Level Measurements Into Traceable Technical Dashboards

github.com/poojakira/mlsec-dashboards (public, default branch main, primary language Python). MIT • Python 3.12 • source snapshot a14ee69 • verified 2026-10-01

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

8	37	FastAPI	token
REPOS AGGREGATED	TEST FUNCTIONS	Dashboard framework	Auth on API

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 **Inspect the trust boundary** →

02 **Apply repository-specific security checks** ↓

03 **Record reproducible evidence**

Research Contribution

An evidence-centered dashboard server that aggregates committed per-repo JSON into traceable HTML + a token-authed REST API, deliberately surfacing weak results (e.g. ROC-AUC 0.54, OOD F1 0.72) alongside strong ones rather than curating highlights.

Verified Evidence

EV-01 Design system: shared

EV-02 Grouped F1 (redteam): 97

EV-03 OOD F1 (redteam): 72

EV-04 Weak ROC-AUC (poison): 54

EV-05 VERIFIED_AT_SNAPSHOT - Aggregates evidence from 8 active ML security repos - README; dashboard_server.py discovers sibling evidence/*.json.

EV-06 VERIFIED_AT_SNAPSHOT - 37 test functions - Counted def test_ definitions in tests/ at source snapshot a14ee69; local verification records 37 passing pytest tests.

EV-07 VERIFIED_AT_SNAPSHOT - Shows weak results explicitly (OOD F1 0.72, ROC-AUC 0.54) - README states weak-alongside-strong design; numbers sourced from sibling evidence files.

EV-08 VERIFIED_AT_SNAPSHOT - Token-authed REST API + static HTML - README features; dashboard_server.py.

EV-09 UNSUPPORTED (disclaimed) - Live SOC monitoring / real-time alerting - README: static evidence snapshots, not live monitoring. Not claimed.

Method and Control Path

01 **Inspect the trust boundary**

02 **Apply repository-specific security checks**

03 **Record reproducible evidence**

Limitations and Residual Risk

- Static evidence snapshots, not live monitoring.
- Data can lag the current CI state of each repo.
- No alerting or real-time telemetry.
- Depends on sibling repos committing evidence.
- Local aggregation; not a hardened service.

Reproducibility

```
Evidence: dashboard_server.py, sibling evidence/*.json,
tests/
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity